EA-0053 - AQELYN Endpoint Security Assessment Engine

Implementation Specification: IS-036

Publication Standard: FULL_COMPLETE

Source of Truth: Master Markdown

Brand: AQELYN

Domain: Endpoint Platform

Architecture Baseline: AQELYN Platform v1.0

gt; This archive is normalized for the AQELYN brand. Historical codenames are intentionally not used in the active implementation baseline.

Section 001 - Document Control

Defines archive identity, publication scope, engineering ownership, and compliance with the AQELYN Architecture Baseline v1.0.

Field	Value
Project	AQELYN
Engineering Archive	EA-0053
Implementation Specification	IS-036
Title	AQELYN Endpoint Security Assessment Engine
Domain	Endpoint Platform
Repository Status	Fixed and immutable
Publication Status	FULL_COMPLETE

Engineering Notes

This section is part of the FULL_COMPLETE implementation baseline. It is intentionally implementation-oriented and shall be used directly by development agents, reviewers, testers, and release managers.

Acceptance Implications

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Section 002 - Executive Summary

Summarizes the role of the engine or standard in AQELYN and explains how it supports customers, companies, and government organizations.

Implementation shall preserve AQELYN architecture rules, user-centered output, evidence-first reasoning, deterministic behavior where practical, and complete traceability to requirements and tests.

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Section 003 - Vision and Purpose

Defines why the capability exists, the problems it solves, and how it contributes to understandable, evidence-based cybersecurity.

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Section 004 - Scope

Defines in-scope and out-of-scope responsibilities so implementation remains focused and traceable.

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Section 005 - Stakeholders and Personas

Identifies private users, SMB administrators, enterprise security teams, developers, auditors, executives, and government operators.

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Section 006 - Functional Requirements

Defines the mandatory capabilities that shall be implemented and verified.

- EA-0053-FR-001: The platform shall support security posture scoring.
- EA-0053-FR-002: The platform shall support misconfiguration detection.
- EA-0053-FR-003: The platform shall support risk prioritization.
- EA-0053-FR-004: The platform shall support user-friendly findings.
- EA-0053-FR-005: The platform shall support endpoint remediation instructions.
- EA-0053-FR-006: The platform shall support policy-driven assessment.
- EA-0053-FR-007: The platform shall provide traceable, evidence-backed operation for this capability.
- EA-0053-FR-008: The platform shall provide traceable, evidence-backed operation for this capability.
- EA-0053-FR-009: The platform shall provide traceable, evidence-backed operation for this capability.
- EA-0053-FR-010: The platform shall provide traceable, evidence-backed operation for this capability.

- EA-0053-FR-011: The platform shall provide traceable, evidence-backed operation for this capability.
- EA-0053-FR-012: The platform shall provide traceable, evidence-backed operation for this capability.

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Section 007 - Non-Functional Requirements

Defines performance, reliability, security, usability, privacy, observability, and maintainability expectations.

- Security: All operations shall enforce least privilege and auditability.
- Usability: Findings shall be understandable by non-experts and actionable by experts.
- Performance: Processing shall support incremental operation and graceful degradation.
- Privacy: Data collection shall be minimized and explainable.
- Reliability: Failures shall be isolated, logged, and recoverable.
- Interoperability: Interfaces shall be versioned and backward compatible where practical.

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Section 008 - Architecture Overview

Describes the internal architecture, dependencies, boundaries, and integration points.

The capability follows the AQELYN architecture pattern: adapter layer, normalization layer, evidence layer, event publication layer, policy evaluation, AI-assisted explanation, workflow/remediation, and reporting. The implementation shall not bypass the object model or event architecture.

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Section 009 - Component Model

Defines major components, services, adapters, repositories, and orchestration responsibilities.

- Connector Adapter
- Normalization Service
- Repository Interface
- Evidence Linker
- Policy Adapter
- AI Explanation Adapter

- Workflow Adapter
- Event Publisher
- Dashboard Provider
- Audit Exporter

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Section 010 - Data Model

Defines canonical objects, identifiers, metadata, evidence links, lifecycle fields, and relationships.

Canonical records shall include:

- `global_id`
- `tenant_id`
- `asset_id`
- `source_system`
- `owner`
- `environment`
- `risk_level`
- `evidence_refs`
- `policy_refs`
- `trust_context`
- `created_at`
- `updated_at`
- `lineage`

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Section 011 - Event Model

Defines events produced and consumed through the AQELYN event architecture.

- `EndpointSecurityAssessmentEngineDiscovered`
- `EndpointSecurityAssessmentEngineUpdated`
- `EndpointSecurityAssessmentEngineAssessed`
- `EndpointSecurityAssessmentEngineFindingCreated`
- `EndpointSecurityAssessmentEngineEvidenceLinked`
- `EndpointSecurityAssessmentEngineRecommendationCreated`
- `EndpointSecurityAssessmentEngineWorkflowRequested`

- `EndpointSecurityAssessmentEnginePolicyViolationDetected`
- `EndpointSecurityAssessmentEngineStatusChanged`
- `EndpointSecurityAssessmentEngineArchived`

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Section 012 - API Model

Defines API boundaries, service contracts, request/response patterns, and versioning requirements.

Operation	Purpose
`create`	Provides controlled access to aqelyn endpoint security assessment engine functions.
`read`	Provides controlled access to aqelyn endpoint security assessment engine functions.
`list`	Provides controlled access to aqelyn endpoint security assessment engine functions.
`search`	Provides controlled access to aqelyn endpoint security assessment engine functions.
`assess`	Provides controlled access to aqelyn endpoint security assessment engine functions.
`explain`	Provides controlled access to aqelyn endpoint security assessment engine functions.
`recommend`	Provides controlled access to aqelyn endpoint security assessment engine functions.
`export`	Provides controlled access to aqelyn endpoint security assessment engine functions.
`archive`	Provides controlled access to aqelyn endpoint security assessment engine functions.

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Section 013 - Security Architecture

Defines authentication, authorization, cryptographic integrity, auditability, privacy, and abuse resistance.

- mutual authentication
- role-based authorization
- signed event records
- evidence integrity
- secure configuration
- secret-free logs

- tenant isolation
- audit export
- rate limiting
- input validation

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Section 014 - Evidence and Explainability

Requires every finding and recommendation to include what happened, why it matters, how it was determined, and what evidence supports it.

Every AQELYN finding shall answer:

- What was found?
- Why does it matter?
- How was it determined?
- What evidence supports it?
- What should the user do next?
- What can AQELYN automate safely?

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Section 015 - Workflow and Remediation

Defines human approval, automated workflow creation, rollback, scheduling, and remediation guidance.

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Section 016 - AI Integration

Defines how AI explanations, recommendations, summarization, and reasoning shall remain evidence-bound and auditable.

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Section 017 - Government and Enterprise Controls

Defines traceability, exportable evidence, role separation, retention, policy mapping, and operational assurance.

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Section 018 - Private User Mode

Defines local-first and privacy-preserving operation where appropriate for personal devices and home users.

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Section 019 - Observability

Defines logging, metrics, tracing, dashboards, alerting, and operational telemetry.

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Section 020 - Performance and Scale

Defines capacity planning, throughput goals, latency targets, scaling rules, and degradation behavior.

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Section 021 - Reliability and Resilience

Defines failure handling, retries, circuit breakers, data recovery, and continuity expectations.

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Section 022 - Testing Strategy

Defines unit, integration, system, security, performance, UX, AI, and acceptance testing.

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Section 023 - Acceptance Criteria

Defines the conditions required before implementation can be considered complete.

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Section 024 - Requirements Matrix

Maps requirement IDs to priority and verification methods.

Requirement	Priority	Verification

EA-0053-FR-001	Mandatory	Unit, integration, and acceptance tests
EA-0053-FR-002	Mandatory	Unit, integration, and acceptance tests
EA-0053-FR-003	Mandatory	Unit, integration, and acceptance tests
EA-0053-FR-004	Mandatory	Unit, integration, and acceptance tests
EA-0053-FR-005	Mandatory	Unit, integration, and acceptance tests
EA-0053-FR-006	Mandatory	Unit, integration, and acceptance tests
EA-0053-FR-007	Mandatory	Unit, integration, and acceptance tests
EA-0053-FR-008	Mandatory	Unit, integration, and acceptance tests
EA-0053-FR-009	Mandatory	Unit, integration, and acceptance tests
EA-0053-FR-010	Mandatory	Unit, integration, and acceptance tests
EA-0053-FR-011	Mandatory	Unit, integration, and acceptance tests
EA-0053-FR-012	Mandatory	Unit, integration, and acceptance tests

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Section 025 - Traceability Matrix

Maps requirements to architecture components, tests, evidence, and implementation artifacts.

Requirement	Component	Test	Evidence
EA-0053-FR-001	Connector	Automated verification	Test report and evidence record
EA-0053-FR-002	Normalizer	Automated verification	Test report and evidence record
EA-0053-FR-003	Repository	Automated verification	Test report and evidence record
EA-0053-FR-004	Evidence Linker	Automated verification	Test report and evidence record
EA-0053-FR-005	Policy Adapter	Automated verification	Test report and evidence record
EA-0053-FR-006	AI Explanation	Automated verification	Test report and evidence record
EA-0053-FR-007	Workflow	Automated verification	Test report and evidence record
EA-0053-FR-008	Event Publisher	Automated verification	Test report and evidence record
EA-0053-FR-009	Dashboard	Automated verification	Test report and evidence record
EA-0053-FR-010	Audit Exporter	Automated verification	Test report and evidence record
EA-0053-FR-011	Security Controls	Automated verification	Test report and evidence record
EA-0053-FR-012	Operational Monitoring	Automated verification	Test report and evidence record

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Section 026 - Engineering Journal

Records decisions, assumptions, constraints, risks, and future extension points.

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Section 027 - Example Artifacts

Provides representative output examples, events, recommendations, evidence packages, and reports.

Example artifacts include JSON event samples, dashboard cards, user-friendly findings, expert detail views, evidence export packages, remediation tickets, and audit-ready reports.

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Section 028 - Implementation Guidance

Defines how Codex, Claude Code, developers, and reviewers should implement the package.

Codex, Claude Code, and human developers shall implement this archive only after reading START_HERE.md, EA-0058, EA-0059, EA-0060, EA-0061, and EA-0062. Code shall be delivered with tests, traceability, and documentation updates.

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Section 029 - Review Checklist

Defines approval checks before coding, merge, release, and operational deployment.

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Section 030 - Publication Manifest

Defines the files included in the FULL_COMPLETE package and the validation rules.

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Final Engineering Review

- Brand normalization to AQELYN: Complete
- Repository hierarchy: Fixed
- Source of truth: Master Markdown
- PDF and HTML generated from master source
- Requirements and traceability included
- Implementation ready